

UST-MIL, Russian Federation

WMO: 310410

Lat: 59.639N

Lon: 133.111E

Elev: 179

StdP: 99.19

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.6	-48.1	-52.1	0.0	-50.4	-49.6	0.0	-48.0	3.5	-31.4	3.1	-30.1	0.3	80	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	29.4	19.1	27.6	18.5	25.5	17.4	20.7	27.2	19.5	25.6	18.3	23.7	1.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.5	23.5	17.3	12.6	22.3	16.2	11.8	21.1	60.9	27.2	56.8	25.6	52.9	24.2	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.6	3.9	3.3	-51.8	32.7	3.3	2.2	-54.1	34.2	-56.1	35.5	-57.9	36.7	-60.3	38.3		
(5)			-51.8	22.7	3.3	2.2	-54.1	24.2	-56.1	25.5	-57.9	26.7	-60.3	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.8	-37.0	-30.7	-17.1	-3.4	7.0	14.7	17.7	14.9	6.0	-6.9	-25.8	-34.9	(6)
(7)		DBStd	20.47	6.80	6.28	7.27	5.54	4.17	3.56	3.80	3.55	4.30	6.37	7.47	7.84	(7)
(8)		HDD10.0	7076	1456	1139	841	402	109	5	1	4	129	523	1074	1391	(8)
(9)		HDD18.3	9606	1714	1373	1099	652	350	118	58	118	368	782	1324	1649	(9)
(10)		CDD10.0	570	0	0	0	0	18	146	241	155	11	0	0	0	(10)
(11)		CDD18.3	59	0	0	0	0	0	9	40	10	0	0	0	0	(11)
(12)		CDH23.3	971	0	0	0	0	13	235	544	177	2	0	0	0	(12)
(13)		CDH26.7	264	0	0	0	0	2	54	176	33	0	0	0	0	(13)
(14)	Wind	WSAvg	1.2	0.6	0.8	1.2	1.6	1.8	1.6	1.4	1.3	1.4	1.3	1.0	0.8	(14)
(15)	Precipitation	PrecAvg	481	19	13	15	24	48	52	76	69	63	50	32	20	(15)
(16)		PrecMax	705	36	27	63	73	74	103	167	172	126	140	77	44	(16)
(17)		PrecMin	340	8	5	3	6	13	20	25	21	19	17	8	7	(17)
(18)		PrecStd	83	7	6	12	13	17	18	28	32	24	24	18	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-16.8	-11.0	5.5	13.0	25.0	30.2	33.2	29.4	21.9	9.3	-0.2	-9.8	(19)
(20)			MCWB	-17.0	-11.8	1.4	6.3	14.4	18.8	21.0	19.7	14.5	4.9	-0.4	-9.9	(20)
(21)		2%	DB	-19.9	-15.0	1.8	10.2	21.3	28.2	30.5	27.5	18.8	6.1	-6.9	-16.9	(21)
(22)			MCWB	-20.1	-15.7	-1.2	4.3	12.1	17.4	19.6	18.9	12.4	3.4	-7.8	-17.3	(22)
(23)		5%	DB	-24.0	-17.7	-1.1	8.3	18.9	26.0	28.5	25.5	16.1	3.6	-11.7	-19.8	(23)
(24)			MCWB	-24.3	-18.2	-3.6	3.1	10.6	16.2	19.3	18.0	11.4	1.7	-12.2	-20.1	(24)
(25)		10%	DB	-26.9	-20.8	-3.7	6.2	16.4	24.0	26.3	23.0	13.8	1.6	-15.2	-23.1	(25)
(26)			MCWB	-27.1	-21.1	-5.7	1.7	9.3	15.4	18.5	16.4	9.6	0.1	-15.6	-23.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.9	-11.8	1.6	6.6	14.9	20.5	22.4	21.4	15.0	5.9	-0.4	-9.9	(27)
(28)			MCDWB	-16.8	-11.1	5.4	12.5	25.0	27.1	30.1	27.0	19.8	8.0	-0.1	-9.8	(28)
(29)		2%	WB	-20.1	-15.5	-1.1	4.9	12.9	18.8	21.2	19.9	13.4	3.9	-7.8	-17.1	(29)
(30)			MCDWB	-19.9	-15.0	1.5	9.7	20.3	26.1	28.2	26.2	17.5	5.8	-7.1	-16.8	(30)
(31)		5%	WB	-24.2	-18.2	-3.5	3.4	11.2	17.3	20.3	18.6	11.9	1.9	-12.1	-20.0	(31)
(32)			MCDWB	-24.0	-17.7	-1.2	7.9	17.1	23.9	26.5	23.8	15.2	3.4	-11.7	-19.8	(32)
(33)		10%	WB	-27.1	-21.2	-5.6	1.8	9.8	16.3	19.2	17.2	10.5	0.3	-15.5	-23.3	(33)
(34)			MCDWB	-26.9	-20.8	-3.7	5.8	15.7	22.4	24.9	21.6	13.2	1.6	-15.2	-23.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	13.4	17.9	14.4	12.9	14.8	13.4	13.3	12.0	9.1	9.4	7.5	(35)
(36)			MCDBR	11.1	14.6	18.0	15.2	18.7	19.7	17.4	17.2	16.4	10.1	11.9	10.0	(36)
(37)			MCWBR	10.8	14.3	15.0	10.3	10.9	9.9	8.3	9.4	10.4	7.4	11.5	9.8	(37)
(38)		5% WB	MCDWB	11.1	14.8	17.0	14.4	16.5	16.1	14.8	14.5	13.9	8.7	12.0	9.8	(38)
(39)			MCWBR	10.9	14.3	14.1	9.9	9.9	9.0	7.7	8.4	9.2	6.6	11.5	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.235	0.271	0.313	0.389	0.455	0.439	0.521	0.458	0.353	0.307	0.245	0.211	(40)
(41)		taud		2.005	2.055	2.033	1.955	2.031	2.192	1.952	2.109	2.390	2.272	2.032	1.938	(41)
(42)		Ebn at Noon		537	721	801	794	769	794	706	726	771	681	509	412	(42)
(43)		Edh at Noon		49	82	115	151	152	132	164	130	83	66	46	34	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.57	1.65	3.39	5.04	5.18	5.90	5.40	4.19	2.67	1.25	0.72	0.32	(44)
(45)		RadStd		0.03	0.07	0.12	0.22	0.42	0.39	0.32	0.34	0.17	0.17	0.06	0.01	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (4 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	-164	N/A	N/A	(47)

Nomenclature: See separate page